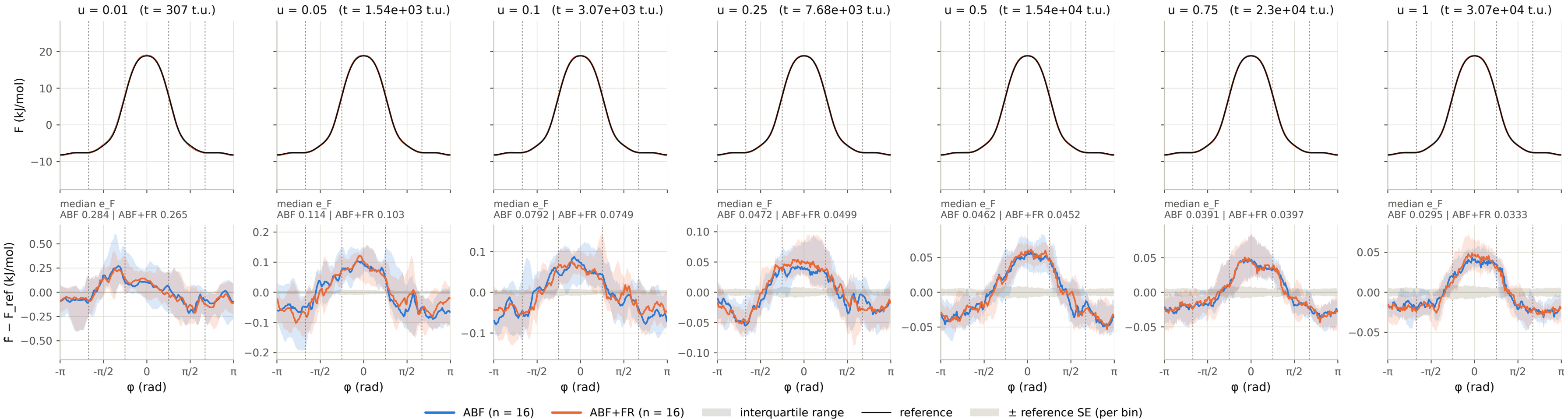


# Ethane/LTA 300 K · N = 2 · T\_N = 30720 t.u.

n\_steps = 153,600,000 · h = 0.0002 · B = N n\_steps = 3.072e+08 walker-steps · complete seeds: ABF 16/16, ABF+FR 16/16

Free-energy profile snapshots  $\bar{F}_t$  vs reference (top) and difference (bottom)



Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC\_PLAN section-4 quantities, unsmoothed.

Profile = histogram\_pmf(Gamma), Gamma from the production accumulators once they hold counts; aligned by removing the mean of ( $\bar{F}_t - F_{ref}$ ) over the circle: the alignment  $e_F$  uses, identical for every method, seed and N. Top: profiles; bottom: difference (y-scale per column).

Dashed guides:  $|\phi| = 0.791$  rad (window  $|z| < 1.5$  Å) and  $|\phi| = 2.109$  rad (cage  $|z| > 4$  Å),  $z = |\phi| a / 2\pi$ ,  $a = 11.919$  Å. Top-row y range = the reference over the scored window  $\pm 35$  % of its span (larger excursions are clipped there and show in the difference row). No smoothing.

eqb\_plot\_config/2 · config lta\_300K\_production.json · 2026-10-10 09:46 UTC